BASAL GANGLIA (NUCLEI)

Prof Yama OE Teaching Notes Series for Medical Students

WHAT ARE THE BASAL GANGLIA?

Think of the basal ganglia as a group of **control centres** buried deep inside the brain. They are made up of **grey matter** (nerve cell bodies) sitting inside the white matter of the cerebral hemisphere, close to the base of the brain.

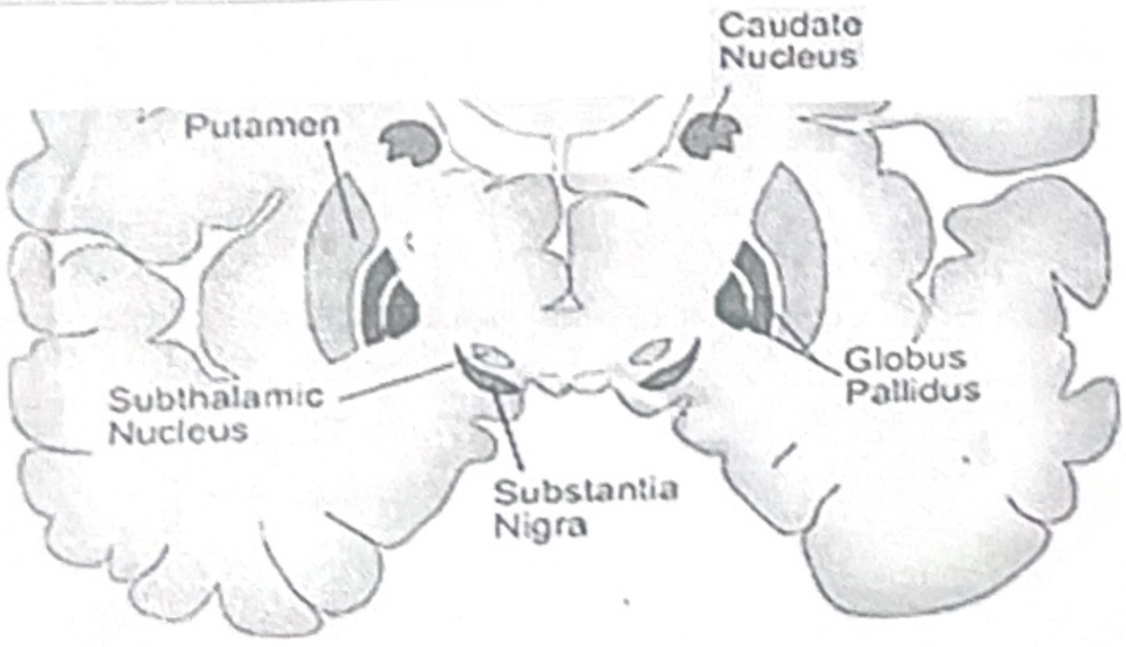
Grey matter	Brain tissue made mostly of nerve cell bodies (neurons). Responsible for processing information.
White matter	Brain tissue made mostly of nerve fibres (axons) covered in a fatty coating called myelin. Responsible for carrying signals between areas.

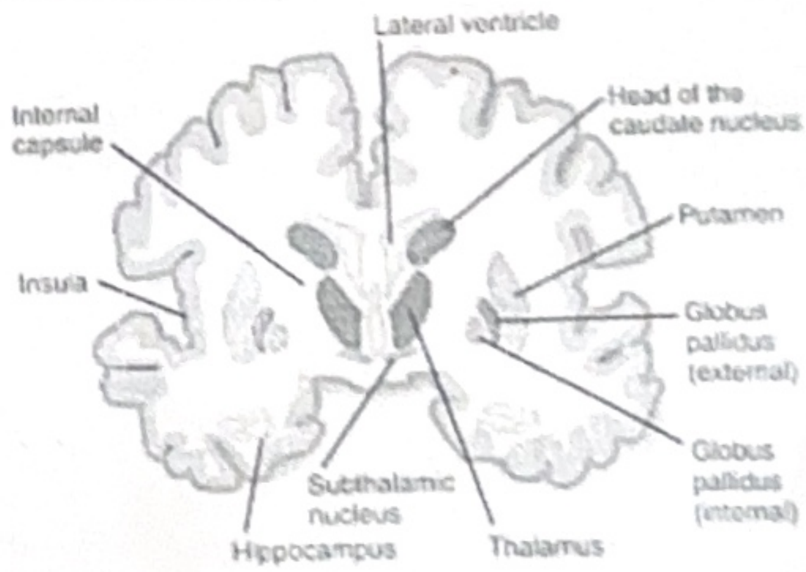
Their main job? They don't start movement — they CONTROL and REFINE it. Think of them as the brain's volume knob for movement — turning it up or down as needed.

THE 5 MEMBERS OF THE BASAL GANGLIA FAMILY

Name	Simple Description
Caudate Nucleus	C-shaped structure; the brain's movement **planner**
Putamen	Sits beside the caudate; helps with motor **control**
Globus Pallidus	The main OUTPUT station — sends instructions **out**
Subthalamic Nucleus	A modulator — **fine-tunes** movement signals
Substantia Nigra	Makes **dopamine**; dark because of melanin pigment

Memory tip: "Can Paul Get Some Naps?" → Caudate → Putamen → Globus pallidus → Subthalamic nucleus → Substantia nigra





HOW THEY CLUSTER TOGETHER

Students often get confused here. Think of it like groups within a group:

1. **Striatum = Caudate + Putamen**
   - These two are best friends — same structure, same function, same origin
   - They are the main ENTRY DOOR — information comes IN here
   - *Memory tip: "Striped CP" — Cats Play*

2. Lentiform Nucleus = Putamen + Globus Pallidus

- Shaped like a lens (biconvex)
- Completely buried inside the brain
- *Memory tip: "Lens-shaped PG"*

Biconvex	Curved outward on both sides — like a lens from a pair of glasses.

3. Corpus Striatum = Caudate + Lentiform Nucleus

- The biggest grouping
- Lies beside the thalamus, partly cut through by the internal capsule
- Serves as a major extrapyramidal motor centre — controls movement outside the main pyramidal system

Extrapyramidal	Movement pathways that do NOT travel through the pyramidal tract. They control automatic background movements like posture, balance, and muscle tone.

Quick Summary: Striatum = C + P | Lentiform = P + G | Corpus striatum = C + (P + G)

THE CAUDATE NUCLEUS — *up close i.e. examining it more carefully*

Picture a comma or the letter C. It has three parts:

- **Head** — the big front part; forms part of the wall of the front horn of the lateral ventricle
- **Body** — arches over the thalamus like a bridge
- **Tail** — curves down and forward, ending at the amygdala

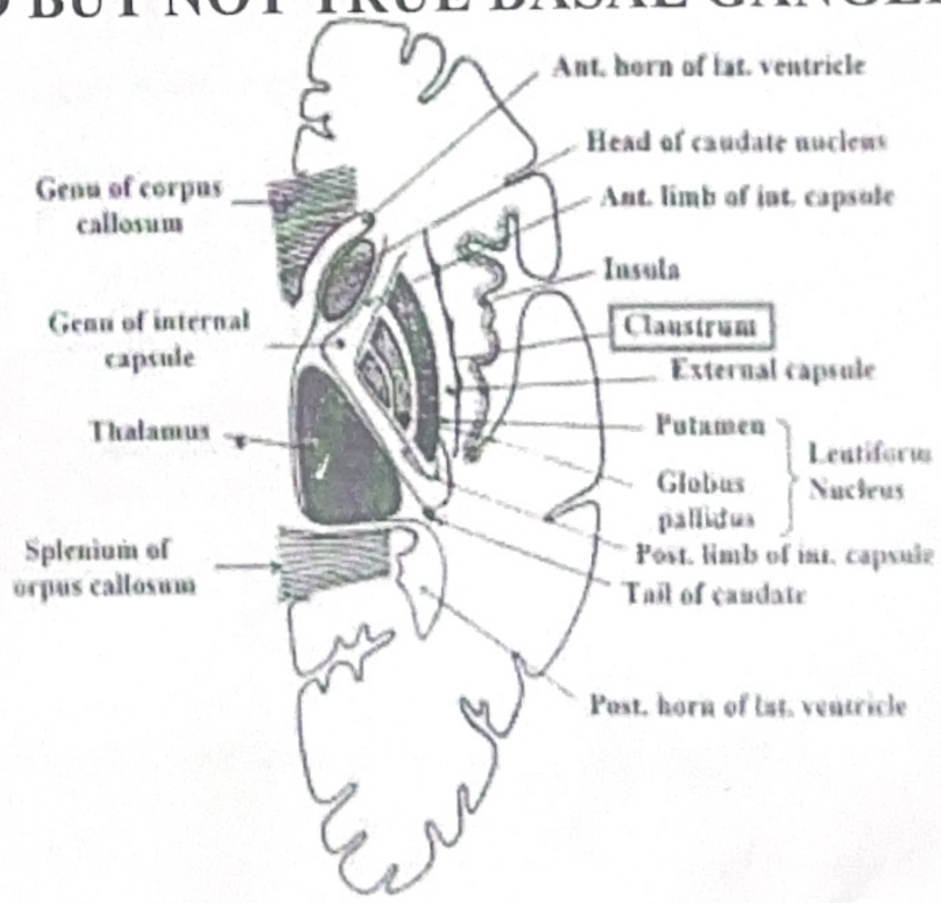
Key point: The head of the caudate is separated from the putamen by the **internal capsule** — but they are still connected below.

Lateral ventricle	A fluid-filled space inside the brain. The caudate nucleus forms part of its wall. The fluid inside is called cerebrospinal fluid (CSF).

TWO STRUCTURES CLOSELY RELATED BUT NOT TRUE BASAL GANGLIA

Claustrum

- A thin sheet of grey matter
- Sits between the lentiform nucleus (medially) and the insula (laterally)
- Separated from lentiform nucleus by the external capsule
- Function: coordinates and integrates activity across different brain areas
- *It is mentioned with basal ganglia because of its location — but its job belongs to cortical integration*

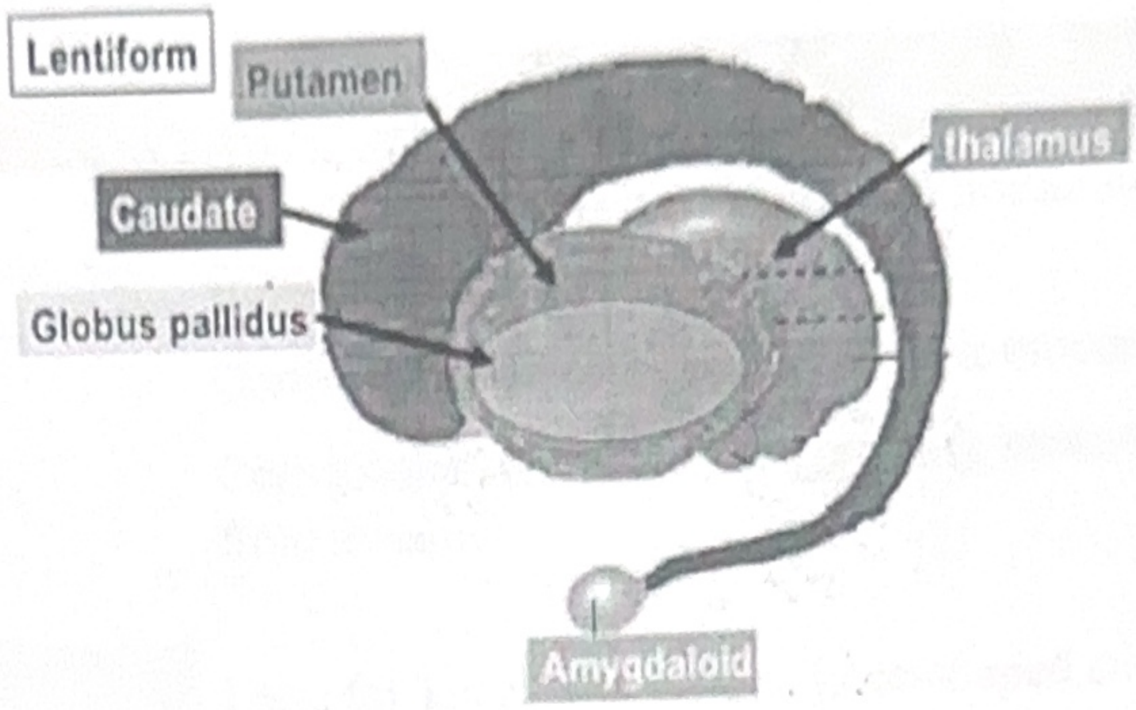


Amygdala (Amygdaloid Nucleus)

- Small oval mass sitting in the uncus, at the end of the caudate's tail

- Part of the limbic and olfactory systems
- Controls emotion, motivation, and smell perception
- *It is mentioned with basal ganglia because of its location — but its job belongs to emotion and the limbic system*

Putamen – lateral medullary lamina – Globus Pallidus
Lat GP – medial medullary lamina – Med GP



Limbic system	A group of brain structures involved in emotion, memory, and motivation. Includes the amygdala and hippocampus.

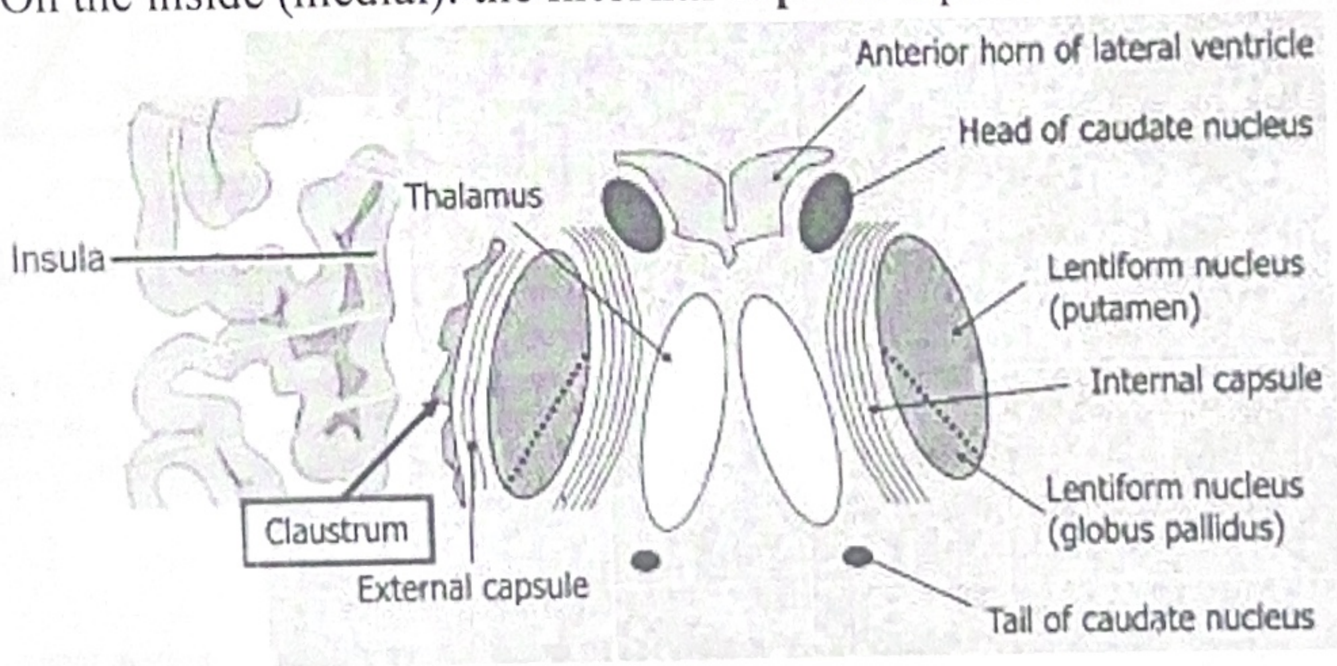
Bottom line: Both structures are neighbours of the basal ganglia — not members of the family.

THE LENTIFORM NUCLEUS — UP CLOSE

- Shaped like a biconvex lens
- **Putamen** — the outer, darker, larger part
- **Globus Pallidus** — the inner, paler, smaller part

Relations — what surrounds it?

- On the outside (lateral): the **external capsule**
- On the inside (medial): the **internal capsule** separates it from the thalamus and caudate



Simple image: *The lentiform nucleus is sandwiched between two capsules — like a filling in a sandwich.*

THE INTERNAL CAPSULE — IMPORTANT NEIGHBOUR

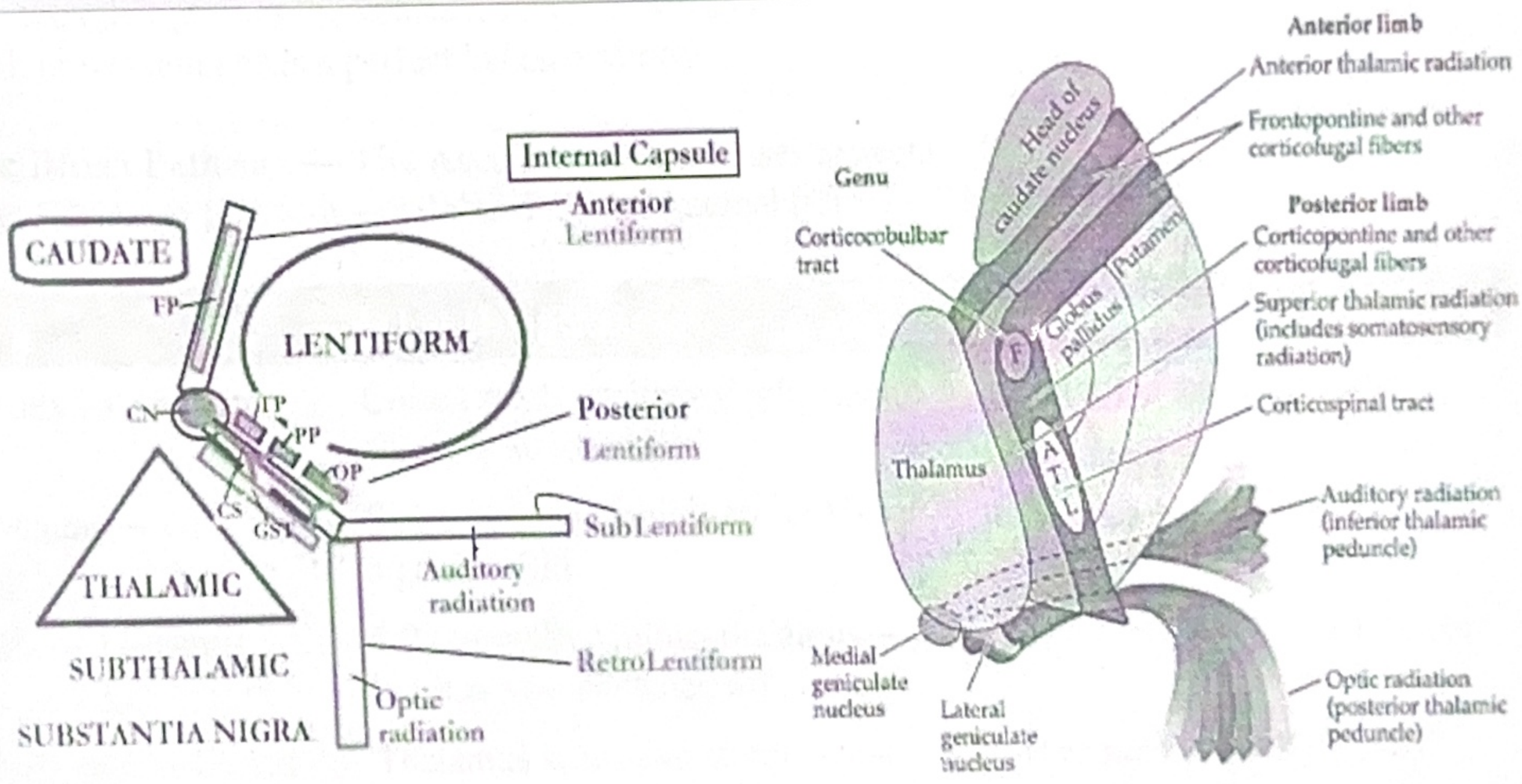
- A broad band of **white matter** (nerve fibres, not nerve cells)
- It is NOT part of the basal ganglia — but it runs right through them
- Carries messages to and from the cerebral cortex

Three parts and what they carry:

Part	What it Carries
Anterior limb	Connections between thalamus and frontal lobe; frontopontine fibres
Genu (the bend)	Corticobulbar fibres (face, head, neck movements)
Posterior limb	Corticospinal tract (body movement); sensory fibres from thalamus

Genu	Latin for 'knee.' The bend or knee-shaped curve in the middle of the internal capsule.

🧠 **Memory trick:** Fugal = away from cortex (motor). Petal = toward cortex (sensory).



Corticonuclei (CN)
CorticoPontine: FrontoPontie (FP), TemporoPontine (TP), ParietoPontine (PP),
OccipitoPontine (OP) General Sensory+Taste (GST) Corticospinal (CS)

SUBSTANTIA NIGRA & SUBTHALAMIC NUCLEUS

Substantia Nigra

- Found in the **midbrain**, between the cerebral crus and tegmentum
- Appears **dark** because its cells contain **melanin pigment**
- **Pars compacta** — makes dopamine
- **Pars reticulata** — sends output signals.....(makes GABA)

- *Think of it as the brain's dopamine factory*

Dopamine	A chemical messenger (neurotransmitter) that mostly has an inhibitory (calming) effect on movement circuits. Low dopamine → Parkinson's disease.

Subthalamic Nucleus

- Sits at the border between the **thalamus** (superiorly/dorsal) and **midbrain** (inferiorly/ventral), behind the hypothalamus.
- Acts as a fine-tuning regulator of movement
- When damaged → causes wild, flinging movements called **Hemiballismus**

★ THE DIRECT & INDIRECT PATHWAYS

📌 **Why this matter:** The connections table tells you what fibres go where, HOW the system actually works. Every exam question about basal ganglia disease links back to these two pathways.

Think of it this way: The basal ganglia run **two (2)** opposite pathways:

a) one presses the **accelerator** (direct pathway),
b) the other presses the **brake** (indirect pathway).

Smooth movement needs a perfect balance of both.

a) The Direct Pathway — The Accelerator (Increases movement)
Route: Cortex → Striatum → Globus Pallidus internal (GPi) → Thalamus → Cortex

Step	What Happens	Effect
1. Cortex → Striatum	Cortex sends excitatory (glutamate) signal to striatum	Striatum is switched ON
2. Striatum → GPi	Striatum sends inhibitory (GABA) signal to GPi	GPi is switched OFF
3. GPi → Thalamus	GPi normally inhibits thalamus — but it is now switched off	Thalamus is FREE to activate
4. Thalamus → Cortex	Thalamus sends excitatory signal back to cortex	Movement is INCREASED

💡 **Simple rule:** Direct pathway = double negative = movement goes UP. Striatum OFF-switches the OFF-switch → thalamus is freed → more movement.

b) The Indirect Pathway — The Brake (Reduces movement)
Route: Cortex → Striatum → Globus Pallidus external (GPe) → Subthalamic Nucleus (STN) → GPi → Thalamus → Cortex

Step	What Happens	Effect
1. Cortex → Striatum	Cortex sends excitatory signal to striatum	Striatum is switched ON
2. Striatum → GPe	Striatum inhibits GPe	GPe is switched OFF
3. GPe → STN	GPe normally inhibits STN — but GPe is now off	STN is freed and becomes MORE active
4. STN → GPi	STN excites GPi	GPi becomes MORE active
5. GPi → Thalamus	GPi now strongly inhibits thalamus	Thalamus is SUPPRESSED
6. Thalamus → Cortex	Thalamus sends less signal to cortex	Movement is DECREASED

Simple rule: Indirect pathway = suppresses movement. The brake pedal. Keeps unwanted movements from happening while you perform a specific action.

How the Two Pathways Work Together

When you decide to move your right hand, the brain does two things at once:

- **Direct pathway** fires → right hand movement is amplified and released
- **Indirect pathway** fires → all other body movements are suppressed

This is how the brain creates focused, purposeful movement — amplifying what you want and suppressing what you don't.

How Dopamine Controls Both Pathways

The substantia nigra releases dopamine into the striatum. This has two opposite effects:

Pathway	Receptor Type	Effect of Dopamine
Direct pathway	D1 receptors	Dopamine EXCITES the direct pathway → MORE movement
Indirect pathway	D2 receptors	Dopamine INHIBITS the indirect pathway → LESS braking → MORE movement

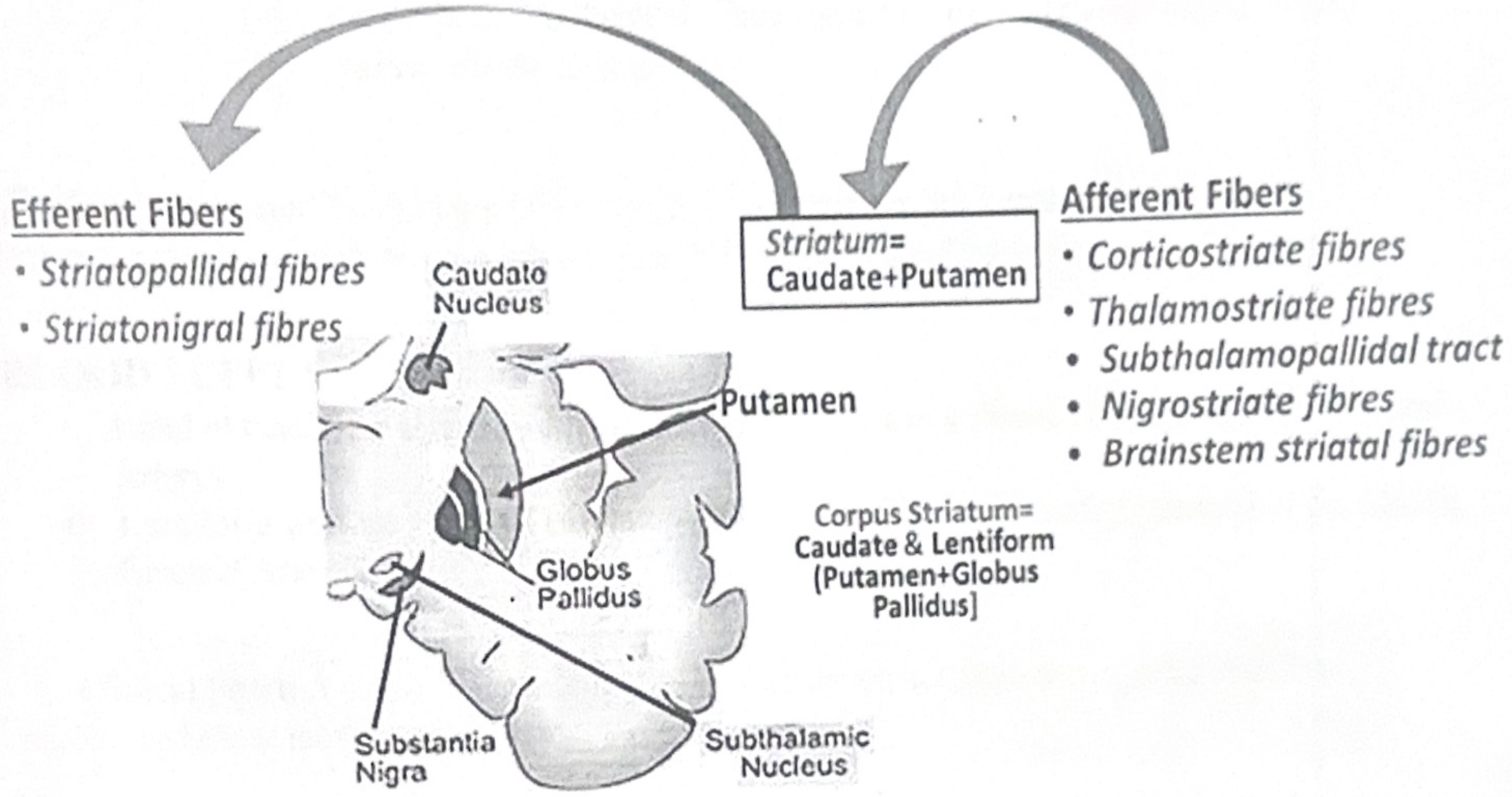
Key Insight: In Parkinson's disease, dopamine is lost. BOTH effects disappear: the accelerator weakens AND the brake becomes overactive. Result: movement becomes slow, stiff, and hard to start.

CONNECTIONS — HOW INFORMATION FLOWS

INFORMATION COMES IN (Afferents) → to the Striatum

Fibre	Where from	Chemical
Corticostriate	Cerebral cortex	Glutamate (excitatory)
Thalamostriate	Thalamus	Glutamate (excitatory)
Subthalamopallidal	Subthalamic nucleus	Glutamate (excitatory)

Fibre	Where from	Chemical
Nigrostriate	Substantia nigra	Dopamine (inhibitory)
Brainstem striatal	Brainstem	Serotonin (inhibitory)



INFORMATION GOES OUT (Efferents) → from the Striatum

Fibre	Where to	Chemical
Striatopallidal	Globus pallidus	GABA (inhibitory)
Striatonigral	Substantia nigra	GABA (inhibitory)

Afferent	Fibres that carry signals INTO a structure. Think: 'Arriving at.'
Efferent	Fibres that carry signals OUT OF a structure. Think: 'Exiting from.'

📌 **Simple rule:** Information enters through the STRIATUM. Information exits through the GLOBUS PALLIDUS.

NEUROTRANSMITTERS

Chemical	Role
Glutamate	Excites — gets things going
Dopamine	Mostly inhibits — calms things down
GABA	Inhibits — applies the brakes
Serotonin	Inhibits — calms movement

Chemical	Role
Acetylcholine	Can do both — depends on the situation

Neurotransmitter	A chemical released at a nerve junction (synapse) to pass signals from one nerve cell to the next. They can either excite (switch on) or inhibit (switch off) the next cell.

🎵 **Memory rhyme:** "GAS to go; DSG says stop." Glutamate and Substance P excite. Dopamine, GABA, and Serotonin inhibit. Acetylcholine can go either way.

BLOOD SUPPLY

- **Head of caudate nucleus** → Medial lenticulostriate artery (branch of the Anterior Cerebral Artery)
- **Lentiform nucleus + rest of caudate** → Lateral lenticulostriate artery (branch of the Middle Cerebral Artery)

Clinical Pearl: A stroke in the middle cerebral artery territory can damage the lentiform nucleus and cause movement problems.

WHAT HAPPENS WHEN THE BASAL GANGLIA ARE DAMAGED?

Think of the basal ganglia as a brake and accelerator system for movement:

- **Too much brake** → movement becomes slow and stiff
- **Too little brake** → movement becomes excessive and uncontrolled

Disease	Lesion Site	What You See
Parkinson's Disease	Substantia Nigra	Tremor at rest (pill-rolling), stiffness (lead pipe rigidity), slow shuffling walk, mask-like face
Chorea (Huntington's, Sydenham's)	Caudate Nucleus	Sudden jerky, purposeless, dance-like movements
Athetosis	Striatum	Slow, writhing, worm-like movements
Hemiballismus	Subthalamic Nucleus	Wild, violent flinging of one arm or leg
Dystonia	Basal ganglia circuits	Sustained twisted body postures

PARKINSON'S DISEASE

The **substantia nigra** stops making enough **dopamine**. Without dopamine, the corpus striatum loses its ability to smooth out movement. The direct pathway weakens AND the indirect pathway (brake) becomes overactive — so movement is slowed and stiffened.

The three classic features — remember TRA: T = Tremor (pill-rolling at rest, stops with movement) | R = Rigidity (stiff like a lead pipe, worse in flexors → stooped posture) | A = Akinesia (slowness and poverty of movement; no arm swing, shuffling gait, quiet monotonous speech, expressionless face)

CHOREA vs HUNTINGTON'S — KNOW THE DIFFERENCE

	Huntington's Chorea	Huntington's Disease
What it means	Just the movement disorder	The FULL picture
Features	Involuntary jerky movements only	Movements + dementia + emotional changes
Cause	CAG repeat on chromosome 4	Same — but full syndrome

What is a CAG repeat? CAG is a sequence of three DNA building blocks (cytosine–adenine–guanine) that codes for the amino acid glutamine. Normally, humans have 6–35 of these repeats in the huntingtin gene on chromosome 4. In Huntington's disease, the repeat number expands abnormally to 36 or more. The more repeats, the earlier the disease appears. This is called ANTICIPATION — each generation tends to inherit more repeats, so the disease starts younger and worsens faster.

Autosomal dominant	A type of inheritance where ONE faulty copy of a gene (from either parent) is enough to cause the disease. Every child of an affected parent has a 50% chance of inheriting the condition.

Types of Chorea to know:

- **Sydenham's** — after rheumatic fever; common in girls and children
- **Huntington's** — hereditary, autosomal dominant
- **Chorea gravidarum** — occurs in pregnancy
- **Pill-induced chorea** — from oral contraceptives (oestrogen effect)

SUMMARY

The basal ganglia work like a well-organised committee that fine-tunes movement:

- The **cortex** sends a movement idea to the **striatum** (input)
- The **striatum** sends signals along the direct pathway (accelerator) AND indirect pathway (brake)
- The **globus pallidus** (output gate) weighs these and sends the result to the thalamus
- The **thalamus** sends the final signal back to the cortex to execute movement
- The **substantia nigra** keeps everything balanced with dopamine (boosting the accelerator and releasing the brake)
- The **subthalamic nucleus** acts as a checker, making sure nothing goes out of control

The ultimate rule: When any part of this committee breaks down — movement becomes either too little (Parkinson's) or too much (chorea, hemiballismus).